

Simulation of Human Survey Responses for Market Research: The Role of Contextualization in LLM-Based Agents

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Abstract

Market and user research rely on surveys to understand customer preferences. Collecting rich human responses is a costly and time-intensive process. This has prompted growing interest in using large language model (LLM)-based agents to simulate human survey responses. A key open question, however, is how different forms of contextual data shape what such agents represent and how well they reflect individual preferences. Based on data collected in a six-week longitudinal study with 13 participants, combining interviews, diary entries, and surveys, we generated eight variants of multiple LLM-based agents that progressively integrate personal contextual information and evaluated their ability to reflect participants' responses to realistic market research questions. Our results show that incorporating personal data can improve alignment with human responses, with contextualized agents achieving similarity scores up to approximately 0.78 compared to around 0.60–0.68 without participant-specific context. The findings indicate that richer contextualization strategies produce measurable but incremental improvements in how well agents approximate individual preferences.



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CCS Concepts

• **Human-centered computing** → **User centered design**; *User studies*.

Keywords

Human-Agent alignment, Market Research, Survey Automation, Generative AI, Large Language Models (LLM), User Modeling

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1 Introduction

Market research surveys with customers play a central role in informing product design, marketing strategy, and innovation management. In practice, however, survey-based methods remain costly and resource-intensive, requiring repeated engagement with human participants. This has motivated growing interest in computational approaches that could support or extend established user and customer research practices [1, 6].

Large language models have attracted recent attention in this context because they can generate coherent answers to survey-style questions. Beyond surface-level fluency, this capability raises a practical question for market research: Can LLM-based agents be used to model how individuals reason about choices, such as purchasing decisions, brand preferences, or reactions to marketing stimuli? Several studies show that LLMs can generate plausible survey responses or approximate aggregate trends [2, 4, 7]. However, such results say little about whether these responses reflect the preferences and reasoning patterns of specific individuals.

In applied market research, understanding behavior often depends on subtle distinctions in how individuals weigh trade-offs, justify choices, or react to alternatives. Simply producing a reasonable answer is insufficient if the underlying decision logic does not correspond to the person being modeled. Without grounding in participant-specific data, an LLM response may appear convincing while reflecting generic patterns derived from training data rather than an individual's purchasing mindset or behavioral tendencies [9, 18]. If LLM-based agents are to reflect individual perspectives rather than generic averages, some form of grounding in personal information becomes unavoidable.

To move beyond generic responses, recent work has explored incorporating personal data into LLM prompts, such as demographic attributes or qualitative material from interviews and diaries [12, 15, 19]. In human-computer interaction, representational techniques such as personas have long been used to make user perspectives explicit in design processes [13], and LLM-based agents extend this tradition by allowing such representations to be generative and interactive. However, introducing personal context is a methodological choice that can fundamentally shape how an agent behaves, as incorporating personal data involves design decisions about how information is structured, summarized, and retrieved, each of which can lead to different behaviors.

Empirical evidence on the effects of different contextualization strategies remains limited. While some studies demonstrate that agents grounded in extensive qualitative data can reproduce individual survey responses with high fidelity [12], other work highlights risks such as stereotyping, overgeneralization, or misleading rationales [1, 9]. Most existing evaluations focus on a single configuration, making it difficult to understand how alternative approaches compare when applied to the same participants and tasks.

To address these questions, we first collected rich, multi-source participant context to enable systematic comparisons of how LLM-based agents behave under different forms of contextualization. We then conducted a six-week longitudinal study with 13 participants, collecting interviews, diary entries, and surveys on purchasing and consumption-related topics. Using this shared data foundation, we compared multiple agent variants that differ in how participant data is incorporated. This design allows us to examine how methodological choices influence an agent's ability to reflect individual preferences under identical tasks.

Our empirical comparison shows that incorporating participant-specific data can improve preference modeling in some cases, but that these effects depend on how personal information is selected, structured, and integrated. In this paper, we present a systematic comparison of agent variants and their performance to inform future market research practices.

2 Method

2.1 Study Design

We recruited 14 participants (8 female, 6 male) through our organization's user research database. Data from one participant was excluded due to inconsistencies, leaving a final sample of 13 participants (8 female, 5 male; ages 19 to 52, $M = 35.2$, $SD = 10.3$) based in Germany, compensated with 250€. Data collection comprised: (1) Interview 1 (90 to 120 min) on routines, attitudes, buying behavior, and finances [12]; (2) 6-day diary study (daily audio reflections) on everyday life, shopping, and banking [5, 8]; (3) Interview 2 (90 to 120 min) reflecting on diary outputs and product preferences; (4) parallel online surveys (2 questionnaires with in total 111 questions, processing time around 2 hours) mirroring interview content to provide structured data in both closed and open formats; and (5) evaluation surveys with market research questionnaires across four domains. Data were collected in German, transcribed using Whisper [14], anonymized, and processed within the European Economic Area.

2.2 System Overview

From the participant data sources, we derived multiple representations of individual context to support agent-based response generation (Figure 1). These include basic demographic attributes, LLM-inferred persona summaries, and expert-style reflections capturing traits, behavioral patterns, motivations, and financial consumer attitudes. In addition, raw participant materials were retained to enable the retrieval of participant-specific statements.

Using this shared data foundation, we implemented a set of agent variants that differ in how participant information is incorporated during response generation. The variants are organized as a progressive sequence that reflects realistic data availability in applied market research settings. For example, demographic attributes are often available through customer relationship management (CRM) systems [11], conference registration, etc., while richer qualitative information may stem from short interviews, feedback sessions, or prior studies. This structure allows us to examine how increasing levels of personal context affect agent behavior under otherwise identical conditions.

At the baseline, the **Random** variant generates responses without any human framing or participant information, effectively answering survey questions without contextual guidance. The **Generic** variant introduces a minimal prompt instructing the model to respond as a human survey participant, relying solely on general language priors. Subsequent variants incrementally introduce participant-specific context, including demographics, persona summaries, retrieved corpus evidence, and reflective interpretations, culminating in a **Fusion** configuration that integrates multiple context layers. All variants use the same backbone model and identical survey questions, differing only in the contextual information provided at generation time (Table 1).

Raw and processed data were indexed in the LanceDB vector database¹ using semantic chunking with Minishlab `pot-ion-base-8M` embeddings [16], as well as OpenAI's `text-embedding-3-large`

¹<https://www.lancedb.com/>

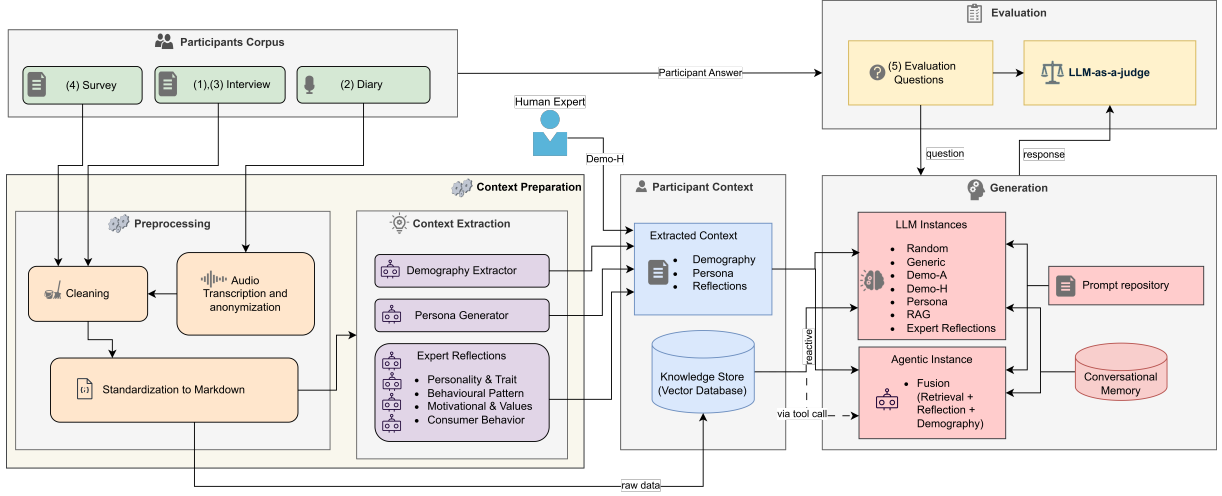


Figure 1: Technical pipeline for agent response generation. Participant data is processed through extraction modules (demographics, personas, reflections), indexed in a vector database, and selectively provided as context to LLM variants. Different agent configurations enable or disable specific context layers.

Agent	Strategy	Context Provided	Source	Method
Random	Direct response generation	None	—	Priors
Generic	Predict as human respondent	None	—	Priors
Demo-H	Human-curated demographics	Age, gender, occupation, income, location	(1), (3)	Manual
Demo-AI	AI-extracted demographics	Age, gender, occupation, income, location	(1)–(3)	LLM
Persona	Rich persona profiles	Demographics, psycho-graphics, values, routines, attitudes	(1)–(3)	LLM
RAG	Retrieved raw data	Top-5 relevant segments (~500 words each)	(1)–(3)	Retrieval
Reflections	Psychological profiling	Big Five traits, behavioral patterns, motivations, financial attitudes	(1)–(3)	LLM
Fusion	Integrated reasoning (CoT)	All combined: demographics, reflections, multi-query retrieval via tool calling	(1)–(4)	Hybrid

Table 1: Agent variant configurations and their context layers. Data sources: (1) Interview 1, (2) Diary study, (3) Interview 2, (4) Surveys. CoT = Chain-of-Thought reasoning.

model². Response generation used GPT-5-2025-08-07 as the backbone model via secure Azure-based infrastructure.

2.3 Evaluation Design

Following established methodologies [3, 10], we employed OpenAI’s O3 model as an LLM-based judge to compare human and agent responses ($n = 1246$ items evaluated), assigning similarity scores from 0.0 to 1.0. To reduce reliance on a single signal, we triangulated these LLM-as-a-judge scores with task-specific metrics (BERTScore, Jaccard, Spearman), finding consistent alignment across formats. We treat these scores as structured similarity indicators rather than definitive ground truth, acknowledging that both metric-based and LLM-based evaluations are proxies for human judgment. Given the exploratory nature of this study and the sample size ($n = 13$), we report confidence intervals for mean differences to convey estimation uncertainty and acknowledge limitations in detecting small effects.

²<https://platform.openai.com/docs/models/text-embedding-3-large>

3 Findings: Incremental Gains from Layered Contextualization

We compared the performance of all agent variants to examine how progressively richer alignment strategies affect similarity with human responses. The Generic baseline served as the reference point for all comparisons, since it reflects the easiest configuration available in practice: an LLM instructed to “act as a human” without any participant-specific data.

Descriptive results in Table 2 and the box plot in Figure 2 show a progression from the lowest-performing Random variant ($\mu = 0.60$) to the highest-performing Fusion method ($\mu = 0.78$). Statistical testing³ confirmed that all contextualized variants achieved significantly higher scores than the baseline. Fusion yielded the largest improvement ($\Delta\mu = +0.092$), followed by RAG ($\Delta\mu = +0.074$) and Persona ($\Delta\mu = +0.071$), with Reflections ($\Delta\mu = +0.059$), Demo-H

³Pairwise comparisons used Welch’s t -test with Benjamini–Hochberg FDR correction. Results were cross-checked with the nonparametric Mann–Whitney U test to account for non-normality. Statistical significance is reported using conventional thresholds. Given the sample size ($n = 13$ participants), these results should be interpreted as exploratory rather than confirmatory.

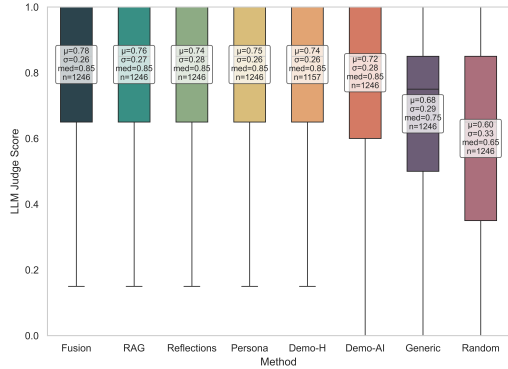


Figure 2: Distribution of LLM Judge similarity scores across methods. Higher scores indicate better alignment with human responses. Fusion shows the highest performance, while Random exhibits the lowest with the greatest variability.

Method	Mean (μ)	$\Delta\mu$	95% CI	Cohen's d	p_{adj}	Sig.
Fusion	0.775	0.092	[0.071, 0.114]	0.336	6.44×10^{-16}	****
RAG	0.757	0.074	[0.052, 0.096]	0.266	1.09×10^{-10}	****
Persona	0.754	0.071	[0.050, 0.093]	0.257	2.98×10^{-10}	****
Reflections	0.742	0.059	[0.036, 0.081]	0.207	3.70×10^{-07}	****
Demo-H	0.736	0.053	[0.030, 0.075]	0.190	3.48×10^{-06}	****
Demo-AI	0.719	0.036	[0.013, 0.058]	0.126	1.68×10^{-03}	**
Generic (Baseline)	0.683	0.000	–	0.000	–	ns
Random	0.600	-0.083	[-0.107, -0.058]	-0.265	1.09×10^{-10}	****

Table 2: Statistical comparison of agent variants against the Generic baseline. Metrics: $\Delta\mu$ = mean difference from baseline; 95% CI = 95% confidence interval for $\Delta\mu$; Cohen's d = standardized effect size; p_{adj} = Benjamini–Hochberg corrected p -values. Significance levels: ** $p < 10^{-4}$, ** $p < 10^{-2}$, ns = not significant. All contextualized methods significantly outperform the baseline.**

($\Delta\mu = +0.053$), and Demo-AI ($\Delta\mu = +0.036$) providing smaller gains (see Table 2 for 95% confidence intervals). These results suggest that progressively richer contextualization is associated with stronger alignment with human responses, though effect sizes remain moderate. The box plot illustrates the distributions of these effects. Fusion achieved the highest median score (med = 0.85) and maintained relatively stable performance across survey items. RAG, Reflections, and Persona showed similarly robust distributions with medians around 0.85. Demo-H (med = 0.85) and Demo-AI (med = 0.85) both demonstrated consistent performance, though with different mean values reflecting their distributional differences. Random exhibited both the lowest median (med = 0.65) and the widest spread, reflecting greater variability when no alignment information is provided.

4 Discussion

Our exploratory findings show that human-agent alignment appears to be incremental. From an HCI perspective, these results highlight contextualization as a design decision, not merely a technical optimization. Each layer (demographics, personas, retrieval,

reflections) contributed a distinct dimension, while Fusion represents a single agent that integrates all layers through automated retrieval and reasoning. Interpreting these results in light of the agent design reveals how each method behaves. Even the Random variant performed at a nontrivial level (0.60), showing that large language models can generate survey responses that partially overlap with human answers purely by relying on their pretraining. The Generic baseline, where the model was simply instructed to “act as a human,” improved to 0.68. This shows that task framing alone provides guidance: the model could infer cues such as language and cultural context (e.g., German financial products) to approximate an “average” respondent.

Adding demographic attributes raised scores further, with differences emerging between human-created (Demo-H: 0.74) and AI-generated (Demo-AI: 0.72) demographic profiles. Demo-H outperformed Demo-AI, indicating that the source of demographic information may influence alignment. Knowing that a respondent is a 45-year-old engineer living in a particular region provides a factual anchor that shifts responses closer to the individual's observed responses. However, extensive demographic data risks creating a stereotype rather than an authentic human-like representation. Personas extended this further, reaching 0.75, by enriching demographics with attitudinal, motivational, and behavioural traits (e.g., risk aversion, structured planning, long-term orientation). These deeper profiles capture aspects of how participants think and decide, not just who they are.

RAG achieved a higher mean (0.76) by grounding answers in evidence from participant data. By retrieving diary entries or interview excerpts, the model could draw on participants' actual expressions. However, this approach requires real-time access to the participant data corpus, which may not always be feasible. In contrast, demographic, persona, and reflection profiles can be pre-computed as abstract summaries, enabling simulation without continuous retrieval. Reflections (0.74) performed at a similar level to Demo-H, as the interpretive psychological profiles they provide may abstract away from lexical overlap. Nonetheless, reflections added insights into *why* participants behave as they do (e.g., often comparing offers before making financial choices).

The Fusion method yielded the strongest results (0.78) by combining all context layers within a single agent. Rather than relying on one context source, Fusion automatically retrieves relevant participant data and applies Chain-of-Thought reasoning to integrate demographic anchors, retrieved corpus evidence, and expert reflections. This unified approach enables the agent to draw on multiple information sources coherently when generating responses.

While our evaluation design employs multiple signals to ensure robustness, we recognize that automated metrics (both LLM based and formulaic) remain proxies for human perception. Our findings characterize directional patterns in hybrid workflows rather than absolute performance benchmarks, as the sample size ($n = 13$) prioritizes interaction depth over broad statistical generalizability. Furthermore, the Fusion variant evaluates the holistic impact of integrated context to maximize ecological validity; this design choice means that individual component effects cannot be fully isolated. Finally, as this study uses the GPT-5 model family, future work should examine whether these patterns generalize across different LLM architectures.

5 Conclusion and Future Work

This exploratory study provides the first evidence that contextualized LLM agents can approximate individual survey responses with incremental improvements: progressively richer context (demographics to personas, retrieval, reflections, and fusion) increases alignment. It should be noted that this incremental improvement comes with increasing complexity on two levels: 1) For agent variants with higher performance, more complex data sets are required that can only be obtained through time-consuming data collection studies – as demonstrated by the presented study, which used a multi-method approach to establish the database for five of the eight agent variants. 2) Generating higher-performing agent models becomes more complex, and they are more time-consuming. Then, when using agents in market research, one has to find a balance between agent performance and complexity.

Future work highlights several key vectors. Because Fusion aggregates multiple components, the contribution of individual layers requires isolation via ablation methods. Validating alignment with human evaluators and scaling to larger, more diverse participant samples will strengthen generalizability. Finally, while our study did not empirically examine ethical risks, synthetic agents may flatten individual nuance or reproduce stereotypes, underscoring the need for consent, transparency, and strict deployment safeguards [17, 18]. Future research should extend beyond whether AI agents can simulate humans, to when and how such simulations add value without compromising authenticity.

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